

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

JOINS AND SET OPERATIONS

2520030098

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Section – 3

CROSS JOIN

Create database:

```
CREATE DATABASE IF NOT EXISTS bookflow2_db;
```

Output:

```
✓ 113 23:09:11 CREATE DATABASE IF NOT EXISTS bookflow2_db
```

Use database:

```
USE bookflow2_db;
```

Output:

```
✓ 114 23:10:42 USE bookflow2_db
```

Create Tables

```
CREATE TABLE class1 (  
    id INT,  
    name VARCHAR(30)  
);  
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```

Output:

```
✓ 115 23:11:37 CREATE TABLE class ( id INT, name VARCHAR(30) )  
✓ 116 23:11:39 CREATE TABLE class_info ( id INT, address VARCHAR(30) )
```

Insert Data:

```

INSERT INTO class VALUES
(1, 'Sushma'),
(2, 'Sowmya'),
(4, 'Mokshagna');
INSERT INTO class_info VALUES
(1, 'DELHI'),
(2, 'MUMBAI'),
(3, 'CHENNAI');

```

Show data:

```
SELECT * FROM class;
```

Output:

	id	name
▶	1	Sushma
	2	Sowmya
	4	Mokshagna

```
SELECT * FROM class_info;
```

Output:

	id	address
▶	1	DELHI
	2	MUMBAI
	3	CHENNAI

Cross Join

```

SELECT *
FROM class
CROSS JOIN class_info;

```

Output:

	id	name	id	address
▶	4	Mokshagna	1	DELHI
	2	Sowmya	1	DELHI
	1	Sushma	1	DELHI
	4	Mokshagna	2	MUMBAI
	2	Sowmya	2	MUMBAI
	1	Sushma	2	MUMBAI
	4	Mokshagna	3	CHENNAI
	2	Sowmya	3	CHENNAI
	1	Sushma	3	CHENNAI

Inner Join

```
SELECT *  
FROM class  
INNER JOIN class_info  
ON class.id = class_info.id;
```

Output:

	id	name	id	address
▶	1	Sushma	1	DELHI
	2	Sowmya	2	MUMBAI

Inner Join

```
SELECT class.name,  
       class_info.address  
FROM class  
INNER JOIN class_info  
ON class.id=class_info.id;
```

Output:

	name	address
▶	Sushma	DELHI
	Sowmya	MUMBAI

Natural Join

```
SELECT *  
FROM class  
NATURAL JOIN class_info;
```

Output:

	id	name	address
▶	1	Sushma	DELHI
	2	Sowmya	MUMBAI

Left Outer Join

```
INSERT INTO class VALUES
(5,'Krishna');
INSERT INTO class_info VALUES
(7,'NOIDA'),
(8,'PANIPAT');
```

Output:

```
125 23:20:10 INSERT INTO class VALUES (5,'Krishna')
126 23:20:13 INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')
```

Left Join

```
SELECT *
FROM class
LEFT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output:

	id	name	id	address
▶	1	Sushma	1	DELHI
	2	Sowmya	2	MUMBAI
	4	Mokshagna	NULL	NULL
	5	Krishna	NULL	NULL

Left Join

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id=class_info.id
WHERE class_info.id IS NULL;
```

Output:

	id	name	id	address
▶	4	Mokshagna	NULL	NULL
	5	Krishna	NULL	NULL

Right Join

```
SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output:

	id	name	id	address
▶	1	Sushma	1	DELHI
	2	Sowmya	2	MUMBAI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Right Join

```
SELECT *  
FROM class  
RIGHT JOIN class_info  
ON class.id=class_info.id  
WHERE class.id IS NULL;
```

Output:

	id	name	id	address
▶	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Union

```
CREATE TABLE first_table(  
id INT,  
name VARCHAR(30)  
);  
CREATE TABLE second_table(  
id INT,  
name VARCHAR(30)  
);  
INSERT INTO first_table VALUES  
(1,'Radha'),  
(2,'Sairam');  
INSERT INTO second_table VALUES  
(2,'Sairam'),  
(3,'Govindha');
```

Output:

```
✓ 132 23:27:38 CREATE TABLE first_table(id INT, name VARCHAR(30))  
✓ 133 23:27:41 CREATE TABLE second_table(id INT, name VARCHAR(30))  
✓ 134 23:27:44 INSERT INTO first_table VALUES (1,'Radha'), (2,'Sairam')  
✓ 135 23:27:46 INSERT INTO second_table VALUES (2,'Sairam'), (3,'Govindha')
```

Union:

```
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

Output:

	id	name
▶	1	Radha
	2	Sairam
	3	Govindha

Union All

```
SELECT * FROM first_table
UNION ALL
SELECT * FROM second_table;
```

Output:

	id	name
▶	1	Radha
	2	Sairam
	2	Sairam
	3	Govindha

Intersect

```
SELECT * FROM first_table
INTERSECT
SELECT * FROM second_table;
```

Output:

	id	name
▶	2	Sairam

Minus

```
SELECT first_table.*
FROM first_table
LEFT JOIN second_table
ON first_table.id=second_table.id
WHERE second_table.id IS NULL;
```

Output:

	id	name
►	1	Radha

Address availability status:

```
SELECT c.id,  
       c.name,  
       CASE  
         WHEN ci.address IS NULL  
         THEN 'Address Missing'  
         ELSE 'Address Available'  
       END AS Status  
FROM class c  
LEFT JOIN class_info ci
```

```
ON c.id=ci.id;
```

Output:

	id	name	Status
►	1	Sushma	Address Available
	2	Sowmya	Address Available
	4	Mokshagna	Address Missing
	5	Krishna	Address Missing